

An algorithm based on K-means for calculating void ratio of solder joint from X-ray image

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Abstract—The voids of solder joints have a very important impact on the reliability of the integrated circuit. X-ray detection technology is commonly used in the inspection of integrated circuit packaging solder joints for its non-destructive characteristics. Accurate and efficient calculation of void ratio (VR) of X-ray images is of great significance for the quantitative assessment of solder joint quality and the study of its correlation with electrical and thermal parameters of devices. It is different to calculate void rate accurately and automatically because of the noise, inconsistent X-ray and overlap by adjacent components. In this paper, an algorithm for calculating void ratio of various solder joint from X-ray image is proposed. The steps of the algorithm including four major steps: preprocessing, morphological processing, void extraction and VR calculation. The core algorithm is K-means clustering which use gray value, eccentricity and area as the features of void. The typical results of VR algorithm on different test sample of X-ray image like LED, connector, power device etc. shows that the algorithm proposed in this paper can accurately mark the position of the void and obtain the accurate void rate of solder joint from the X-ray image.

Keywords—Solder joint, Void ration, X-ray image, Algorithm

I. INTRODUCTION

The void of solder joints is a common deflection of integrated circuit. Void ratio is an important parameter to measure the quality of solder joints, which is closely related to the thermal resistance, resistance and other parameters of devices[1]. Void in the solder joint is mainly cause by the gas failed to escape at high temperature during soldering process[2]. X-ray detection technology is commonly used in the detection of integrated circuit packaging solder joints for its non-destructive characteristics, It can easily detect the defects hidden in the solder joint such as void, crack, short, open and so on[3-4]. X-ray inspection is divided into two-dimensional (2D Xray) and three-dimensional (3D Xray). For components with relatively simple packaging, 2D Xray inspection method can effectively and quickly detect defects in them. 3D X-ray inspection is better for components with complex internal structures.

Since the void ratio has a very important influence on the reliability of the device, the relevant standards give specific regulations. For example, IPC-7095B specifies acceptance criteria for BGA solder balls. The total area of the cavities cannot exceed 20% of the area of the solder ball when there are multiple cavities in the BGA solder ball, It means that it deemed the acceptance qualified when the void rate does not

exceed 20%. When there is only a single void inside the solder ball, if the void area does not exceed 25% of the total area of the solder balls, it will be considered as qualified, otherwise it will be regarded as unqualified[5]. Therefore, it is very important to accurately and efficiently identify the void and calculate the void ration. Many of algorithms of VR calculation are proposed. ZHANG Jun-sheng proposed a dynamic threshold segmentation to extract the solder ball area and void areas respectively to solve the problem of low grayscale contrast of void defects in X-ray inspection of BGA solder joints[6]. ZHANG also realized an extraction and automatic analysis of void defects in solder joints based on mathematical morphology[7]. Li Jingyuan et al. proposed a new method based on coordinate transformation, which converts the edge of the solder ball from a curve to a straight line and then extracts the edge of the solder ball through non-parametric edge detection, which has stronger adaptability in complex backgrounds[8].

However, the above methods are all aimed at the detection of voids of BGA solder ball. The shape of the solder ball is a circle, and most of its internal voids are regular circles or ellipses for BGA. The applicability of the above method has not been verified for different types of detection objects and irregular-shaped void detection. There are many types of components on the printed circuit board (PCB), and various components around the solder joints may interfere with the identification of the voids, such as voids may be partially blocked. In addition, the density of components on the PCB continues to increase, resulting in a complex X-ray image background. A method such as global or dynamic threshold segmentation cannot effectively detect and extract different components and different voids. So the void detection and void ratio calculation method based on K-means clustering of multi features is proposed in this paper, It can effectively and accurately detect void of different components and voids of different shapes.

II. VR CALCULATION ALGORITHM

A. K-means algorithm

The K-means algorithm is an unsupervised clustering algorithm. The K-means algorithm is one of the most widely used algorithms in cluster analysis. It divides n objects into k clusters according to their attributes so that the obtained clusters satisfy: the objects in the same cluster are more similar; the objects in different clusters are less similar. The algorithm

is simple, easy to understand and implement; it has the advantages of fast convergence and high efficiency.

Given sample set $D=\{x_1, x_2, \dots, x_m\}$, minimize squared error of K-means algorithm for clustering obtained by clustering $C=\{C_1, C_2, \dots, C_k\}$ can be expressed by equation (1).

$$E = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|_2^2 \quad (1)$$

$\mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x$ is the mean vector of cluster C_i [9].

B. Feature of void

The characterization and extraction of void features are very important to accurately extract the void and calculate the void rate for the K-means clustering algorithm. Therefore, it combines the void growth mechanism and the principle of X-ray detection to analyze the void features in this paper. Three types of characteristics of voids including gray value, void shape and void area used in this paper are analyzed.

a) Gray value

Due to the different density and thickness of the devices and solder joints on the PCB, the amount of X-ray absorbed by various parts of the component is different. It causes the change of X-rays reaching the flat-panel detector, so images with different contrasts of black and white are formed on the detector, and the final image is a digital image with different gray values. Due to the existence of voids in the solder joint, the thickness of the solder is thinner than other locations, so in the X-ray pattern, The gray value at the position of the void is larger than the position without void in the X-ray image, but due to noise and blocked by adjacent component. The gray value of the void in the solder joint may be smaller or larger than that of the solder joint, and it cannot effectively identify the void by the gray value alone.

b) Shape of void

From the analysis of the mechanism of void formation, the reason for void existed in the solder joint is because the gas cannot escape in time during the melting process of the solder re-flow. Due to the diffusion characteristics of the gas, the shape of the void is generally a circle or an ellipse, and its boundary is an arc or a curve; it cannot be a rectangle, etc. or a straight line of boundary. The eccentricity can be used to characterize the shape of the void.

c) Area

The voids are located inside the solder joints, which appear as large voids or random small voids, but their area is smaller than that of the solder joints, so the area can be used as one of feature.

C. VR calculation flow

The algorithm flow of the void ratio calculation proposed in this paper is shown in Fig. 1. The algorithm is mainly divided into four steps: preprocessing, Morphologic processing, void extraction, void ratio calculation and void location marking. The input of the algorithm is an X-ray digital image of the device under test, and the output is an X-ray digital image marking the void location and void ratio.

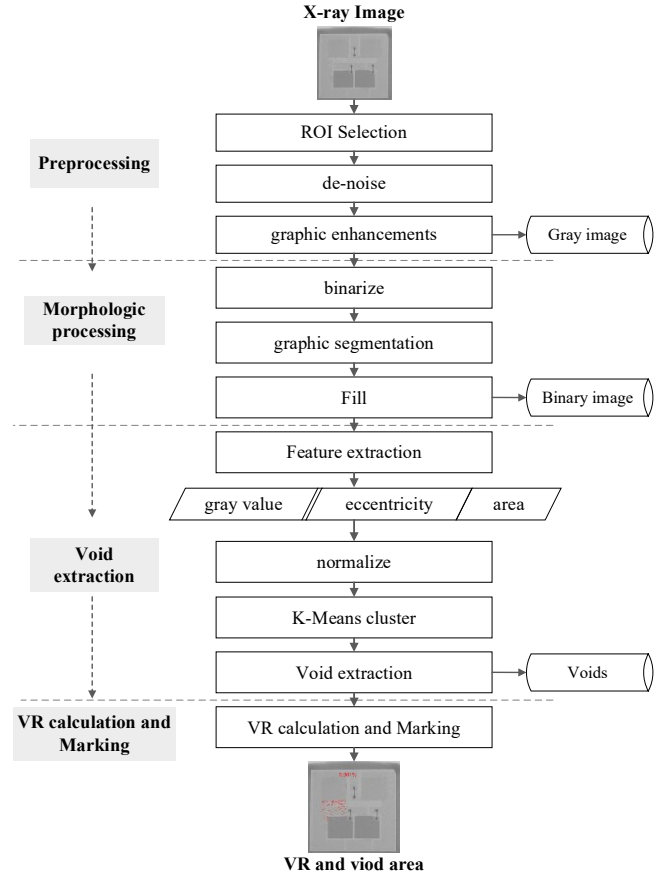


Fig. 1. Void calculation algorithm flow

a) Preprocessing

The input of the pre-processing is the original X-ray digital image of the device under test. There may be many components in the original image. Therefore, it is necessary to select the solder joint(ROI) to be analyzed firstly. The calculation of the void ratio is according to the solder joint to be analyzed. The median filter is used to denoise the image secondly which has a good effect on the removal of speckle noise and salt-and-pepper noise as part of the X-ray imagination procedure. It can reduce noise while keeping the edges sharp. Finally, the image is enhanced based on gray balance.

b) Morphologic processing

The graphics processing mainly segment the image and extracts the position of the void. The image is binarized by adaptive threshold firstly, then the contours of void is obtained by active contour image segmentation using Chan-Vese over 5 iterations, the void is filled lastly.

c) Feature extraction

Three features of void are extracted as gray value, eccentricity and area.

- Feature 1:gray value

The gray value is obtained from the enhanced X-ray image, denoted as g , and for an $M \times N$ image, the range of gray value g for each pixel is $[0, 255]$. The larger the gray value, the closer it is to white on the X-ray image.

- Feature 2: Eccentricity

Eccentricity of the ellipse that has the same second-moments as the region. The eccentricity is the ratio of the distance between the foci of the ellipse and its major axis length expressed as equation (2). The value is between 0 and 1. An ellipse whose eccentricity is 0 is actually a circle, while an ellipse whose eccentricity is 1 is a line segment.

Where e is the eccentricity, c is focal length of an ellipse, a is the length of the major axis of the ellipse.

- Feature 3: Area

The area includes the area of each part of the image after segmentation, expressed in the number of pixels.

d) Normalized

The ranges of each feature's value need to be consistent for the K-means clustering algorithm, otherwise the results will be inaccurate. The features are normalized to the same range of [0,1] by the following equation.

$$\bar{x} = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (3)$$

Where x is the original value of feature. $\min(x)$ and $\max(x)$ represents the minimum and maximum of the feature separately. $\bar{x}$ is the normalized value.

Additionally, since eccentricity and area are attributes of regions, the number of regions and the number of pixels represented by gray values are not balanced, therefore, the pixels belonging to the same region are assigned the same eccentricity and area for balance. For example, it is assumed that the image is divided into L regions of different sizes. All pixels in region k ($k \in [1, L]$) have the same eccentricity and area, but with different gray values.

Then, the feature of each pixel point in the $M \times N$ X-ray digital image can be expressed as the following equation and its dimension is 3:

$$x_i^{(j)} = [\bar{g}_i^{(j)}; \bar{e}_i^{(j)}; \bar{s}_i^{(j)}] \in R^3 \quad (4)$$

Where $x_i^{(j)}$ represents pixels with column i and row j , $i \in [1, M]$, $j \in [1, N]$. $\bar{g}$, $\bar{e}$, $\bar{s}$ are the normalized gray value, eccentricity and area separately

D. K-means cluster

K-means clustering whose number of clusters is 2 is performed with 3 dimensional features. The algorithm marks each pixel as a void or not according to the above characteristics.

E. VR calculation and void Marking

The VR is the ratio of the void area (sum of pixels) in the ROI to the area of the ROI (sum of pixels), expressed as follows:

$$VR = \frac{\sum_{k=1}^L S_k}{S_{ROI}} \quad (5)$$

Where L is the sum of number of void, $\sum_{k=1}^L S_k$ represents the sum area of all void, S_k is the area of k -th of void ($k \in [1, L]$). S_{ROI} is the area of ROI.

III. EXPERIMENTS AND DISCUSSIONS

A. X-ray image description

The X-ray images analyzed in this paper are from DAGE XD7600NT Ruby FP X-ray (2-dimensional CT) and YXLON Y. CT Precision (3-dimensional CT) system. Tested samples include connectors, LEDs, integrated circuits, power devices, and PCBs et. Typical images are shown in Fig. 2.

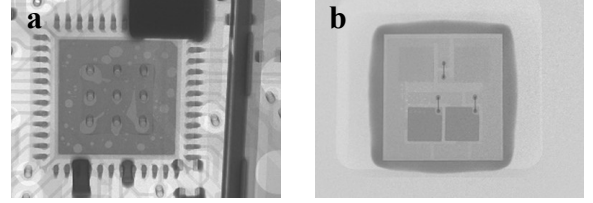


Fig. 2. Typical X-ray image (a) Integrated Circuit (b) Power device

B. Experiment results

The integrated circuit is selected as the typical analysis instance to verify the effective of algorithm because of its complex background of X-ray image shown in Fig. 3. It can be seen there are voids hidden by other components (such as at location ②), the voids are irregular in shape and different in size (such as at location ①③), so it brings the much challenge.

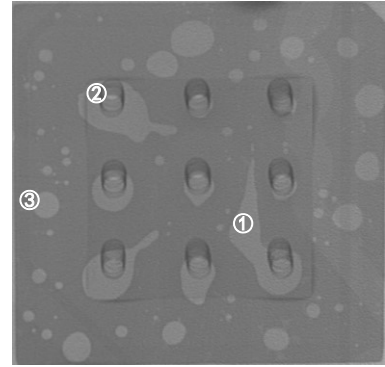


Fig. 3. X-ray image after ROI selection

It is pre-processed according to the image pre-processing process described in Section 2, and the result after image filtering and enhancement is as Fig. 4.

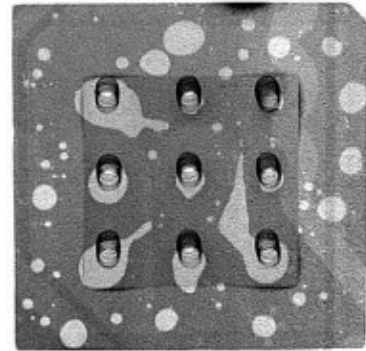


Fig. 4. X-ray image after pre-processing

Then, image processing and void extraction based on K-means are used to calculate the void rate and mark the voids in the original image. The result is shown in Fig. 5. It can be seen from the figure that all the voids are correctly marked, and the effect of shading can also be eliminated for the voids in the masked position of the components, which shows that

the algorithm proposed in this paper has a good effect on the recognition of voids in complex background images.

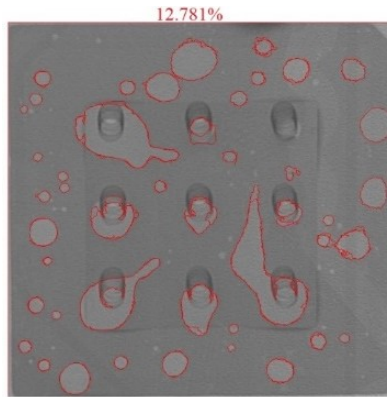


Fig. 5. Caculation results for integrated circuit

Fig. 6 shows the comparison of the results of void extraction using only one feature(gray value) and multidimensional features proposed in this paper(gray value, eccentricity, and area). It can be seen that using the grays value feature alone will identify the imaging background noise as a void, and cannot eliminate the shading of the components. It is not suitable for X-ray image with complex background. Otherwise, the K-means clustering algorithm using 3-dimensional features can more accurately to extract voids.

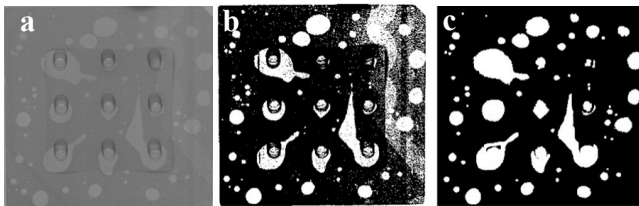
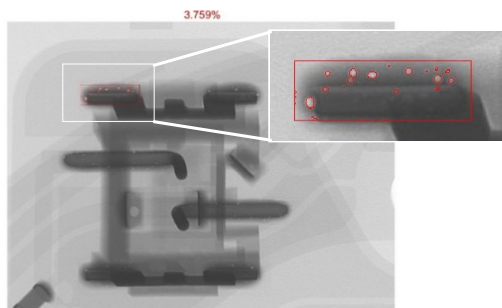
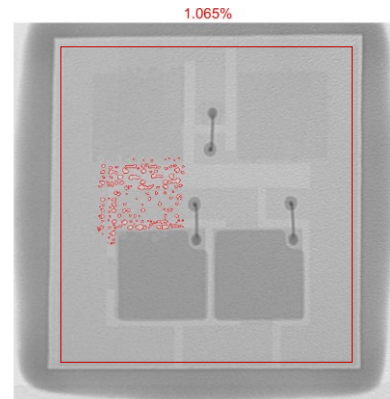


Fig. 6. Comparison of void extraction (a)Original image (b) K-meas with gray value (c) K-meas with 3 features

The algorithm proposed in this paper is applied to other types of component detection. The results of the algorithm proposed in this paper in the detection of voids in connectors and power devices shows in Fig. 7. The red rectangle in the figure is the solder joint area to be analyzed. It can be seen that the algorithm proposed in this paper has a good detection effect for different types of tested samples and voids of different shapes.



(a) Results of some connector



(b) Results of some power device

Fig. 7. Caculation results of a conector and a power device

IV. CONCLUSION

In this paper, an automatic void identification and void ratio calculation algorithm based on K-means clustering with multidimensional features of void is proposed. According to the experiment results of different types of test samples, different image backgrounds, and voids of different shapes and sizes, the algorithm can accurately identify the voids and obtain the void rate. The algorithm is simple to implement and has good robustness.

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